

Online Education & Course **Management System (Educa)** **Project Report**

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Acknowledgement

The project "Online Education & Course Management System (Educa)" is the project work carried out

Name	Enrollment No
Mohan Panthagani	AF05213254

Under the Guidance

I express my gratitude to my project guide and faculty members for providing assistance, suggestions, and technical guidance throughout the design and execution of this web platform.

I am also thankful to my family members, peers, and friends who provided continuous encouragement and support during the development and documentation stages of this project.

Abstract

The Online Education & Course Management System (Educa) is a responsive web application engineered to bridge instructors (tutors) and students (learners) through structured digital learning environments.

The platform eliminates the complexities of manual course distribution by allowing instructors to organize tutorials into modular playlists, upload educational video lectures, and monitor student engagement.

Students can register, browse course catalogs categorized by domain (e.g., Development, Business, Design, Science), stream video content directly within the browser, save playlists to their personal bookmarks, like videos, and submit real-time comments on specific lectures.

The system incorporates cookie-based session authentication, input sanitization routines, and secure PDO prepared statements to safeguard database transactions. Built using PHP, MySQL, JavaScript, and CSS3, the system ensures high usability, fast media delivery, and reliable content management across desktop and mobile screens

1. Introduction

The rapid evolution of Information and Communication Technology (ICT) has fundamentally reshaped academic and vocational learning paradigms. Traditional classroom settings often encounter limitations related to geographical boundaries, inflexible scheduling, and static instructional materials. In contrast, modern web-based learning management systems enable learners to access specialized, self-paced technical courses from anywhere at any time.

The **Online Education & Course Management System (Educa)** is an interactive, full-stack e-learning web platform engineered to bridge the gap between educational content creators (tutors/instructors) and learners (students). Designed with a modular architecture, Educa streamlines the digital distribution of video tutorials, organizes curricular content into structured playlists, and fosters community engagement through integrated feedback and tracking systems.

- **Curriculum & Content Management for Tutors:**

Educators require intuitive tools to structure multi-part lessons without managing complex server infrastructure. Educa provides an administrative environment where verified instructors can create accounts, manage professional bios, organize lectures into topical playlists, and publish MP4 video content along with custom thumbnails and comprehensive descriptions. Visibility controls (active vs. deactive) allow instructors to draft and review coursework before releasing it to students.

- **Discovery & Personalized Dashboards for Students:**

Learners can discover tutorials through category-based navigation (such as Web Development, Business, Design, Software, and Science) and topic tags. Once registered and authenticated via persistent session cookies, students access a personalized dashboard that summarizes their learning activity, including total liked videos, comments made, and saved course bookmarks.

1.1 Objective of the Present Work

The objectives of this project are as follows:

- **Digital Course Distribution & Student Interaction:** Develop a lightweight, scalable web application using a standard PHP and MySQL architecture to facilitate online learning and communication.
- **Structured, Playlist-Based Course Organization:** Group individual video lectures into coherent, modular playlists categorized by domain topics (such as Development, Business, Design, and Science).
- **Role-Based Workflows:** Establish distinct operational interfaces separating student exploration and video streaming from tutor/admin curriculum authoring and media management.
- **Student Engagement Mechanisms:** Implement interactive features allowing learners to like video content, bookmark playlists for future viewing, and participate in discussion threads through editable comments.
- **Data Security & Integrity:** Enforce robust security practices by using PHP Data Objects (PDO) prepared statements with parameter binding and `filter_var()` sanitization routines to prevent SQL injection and cross-site scripting vulnerabilities.
- **Cross-Platform Responsive Design:** Design fluid, accessible user interfaces with HTML5, CSS3, and modern viewport standards to ensure usability across desktop monitors, tablets, and mobile devices

2. System analysis

2.1 Problem Definition

Traditional educational setups and disjointed online video channels lack centralized interaction tracking, organized playlist bookmarking, and dedicated instructor-to-student feedback channels. Furthermore, many existing web platforms suffer from heavy frameworks that introduce overhead for lightweight course streaming.

This project addresses these gaps by delivering an independent, database-driven PHP/MySQL architecture where instructors have full CRUD controls over their curriculum and students have an integrated interface to discover, track, and review educational media.

2.2 Feasibility Study

- **Technical Feasibility:** Developed using PHP (PDO), MySQL, Apache, HTML5, and CSS3. These open-source technologies are mature, cost-effective, and fully compatible across standard hosting environments.
- **Economic Feasibility:** Built entirely on open-source web technologies (XAMPP stack, Font Awesome), eliminating external licensing expenses.
- **Operational Feasibility:** The clean UI layout ensures intuitive navigation for both non-technical students and instructors.
- **Schedule Feasibility:** The modular development architecture enables incremental feature development across defined development sprints.

2.3 Software Requirement Specification (SRS)

Software Requirements:

- **Operating System:** Windows 10 / 11, Linux (Ubuntu/Debian)
- **Web Server:** Apache (via XAMPP / WampServer / Native Apache)
- **Backend Engine:** PHP 7.4 / 8.x (with pdo_mysql enabled)
- **Database:** MySQL 5.7+ / MariaDB 10.4+
- **Frontend:** HTML5, CSS3, JavaScript (ES6+), Font Awesome CDN

Hardware Requirements:

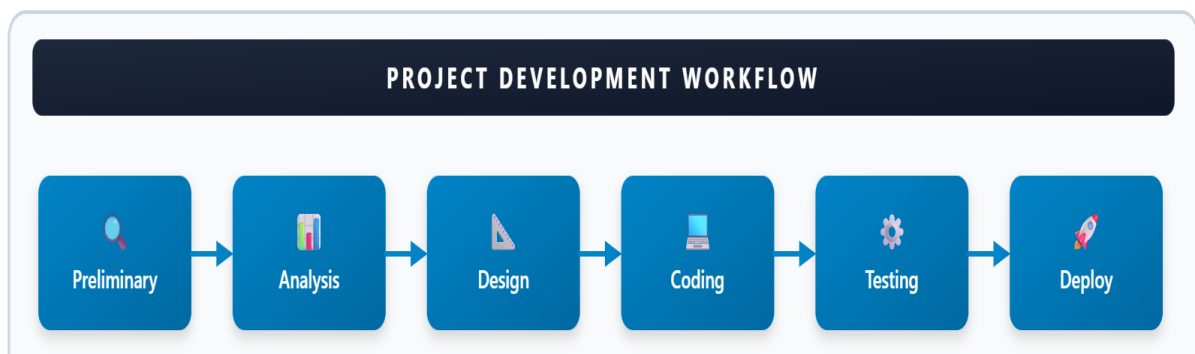
- **Processor:** Intel Core i3 or higher (or equivalent AMD processor)
- **RAM:** Minimum 4 GB (8 GB recommended for local server hosting)
- **Hard Disk:** 500 MB free storage space for codebase and media directories
- **Display Resolution:** 1366 × 768 minimum resolution

2.4 Project Planning

Project planning establishes a structured roadmap for developing the **Online Education & Course Management System (Educa)**. It defines the development lifecycle, resource allocation, and milestones necessary to deliver a reliable web platform.

Phases Covered in the Plan

- **1. Preliminary Investigation:** Evaluating project scope, feasibility, problem definitions, and educational objectives.
- **2. System Analysis:** Gathering functional specifications, analyzing user/tutor workflows, and defining software/hardware requirements.
- **3. System Design:** Architecting database schemas (course_db), Entity-Relationship Diagrams (ERD), Data Flow Diagrams (DFD), and user interface mockups.
- **4. Implementation (Coding):** Developing backend logic in PHP (PDO), configuring the MySQL database, and creating responsive frontend interfaces with HTML5, CSS3, and JavaScript.
- **5. Security Engineering:** Integrating filter_var() sanitization routines, PDO prepared statements, and cookie-based authentication controls.
- **6. Testing & Debugging:** Conducting unit testing, module validation, database integration checks, and cross-device user acceptance testing.
- **7. Deployment & Maintenance:** Deploying the codebase to local/live web servers (Apache) and ensuring continuous operational support.



2.5 Functional Requirements

1. User Module (Learners / Students):

- **Registration & Secure Login:** Create a new account with profile image upload and log in using email/password authentication (SHA-1 encrypted) with persistent cookie storage (user_id).
- **Course & Playlist Exploration:** Browse the entire course catalog, search courses by title or topic, and filter tutorials by category (Development, Business, Design, Science, etc.).
- **Video Streaming:** View structured video playlists and watch individual video lessons in a built-in media player.
- **Interactive Feedback & Comments:** Submit comments on lecture videos, edit existing comments, or delete them in real time.
- **Engagement Tools:** Save complete playlists to personal bookmarks and like specific video lessons.
- **User Dashboard:** View total likes, total comments made, and saved bookmarks from a centralized dashboard.

2. Admin Module (Master Administrator):

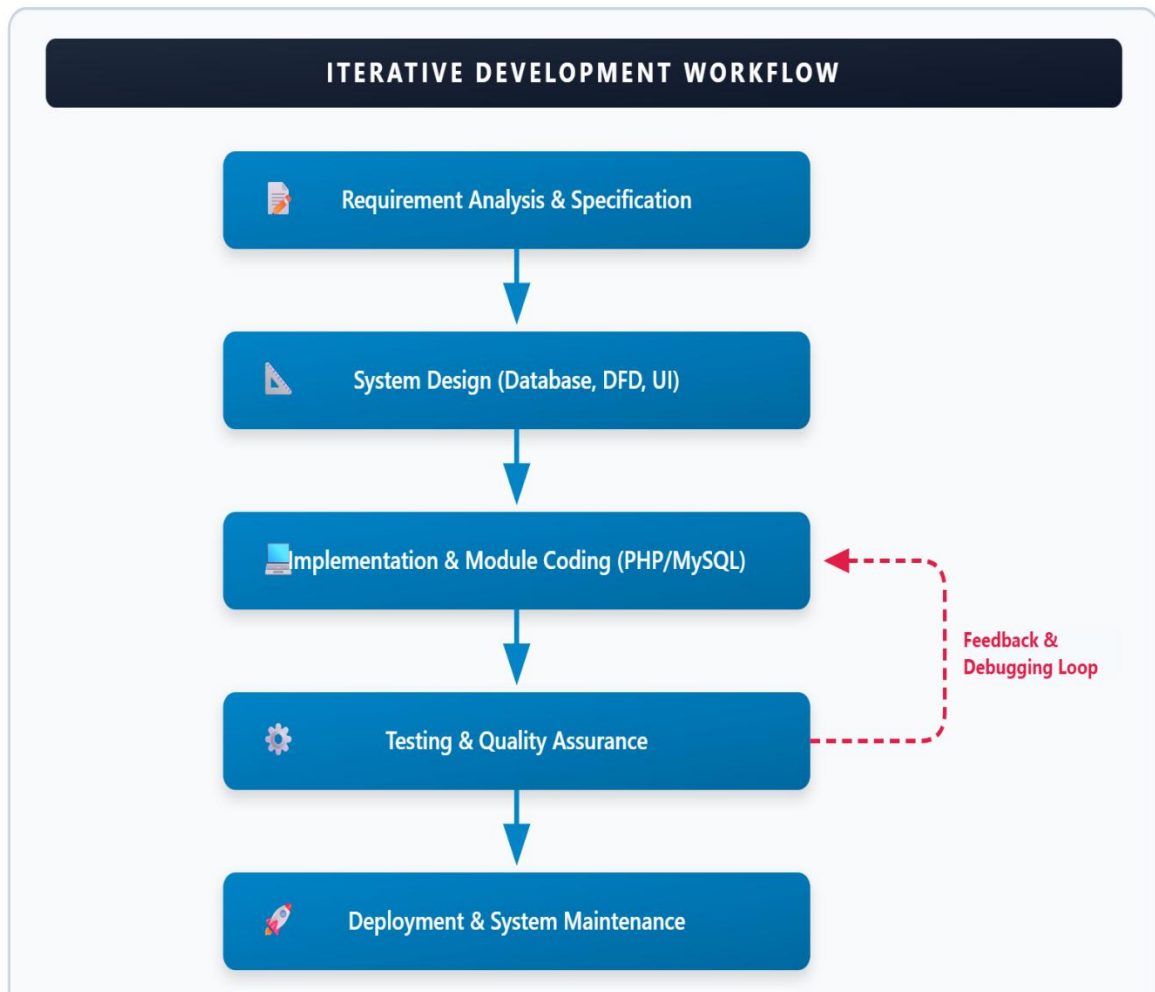
- **Platform Overview:** View real-time platform statistics including total active courses, enrolled students, registered tutors, and contact queries.
- **Instructor Moderation:** Manage registered tutor profiles and review account privileges.
- **Content Governance:** Moderate course playlists and video uploads across the platform to ensure policy compliance.
- **Inquiry Management:** Access, review, and resolve contact messages submitted via contact.php.

3. Sub-Admin Module (Tutors / Instructors):

- **Tutor Authentication:** Register and log in via the dedicated instructor portal (admin/register.php / admin/login.php).
- **Profile Customization:** Manage professional credentials, job title/profession, bio, and profile photo.
- **Playlist Authoring:** Create, edit, and delete structured course playlists with titles, descriptions, and thumbnail banners.
- **Video Management:** Upload MP4 video lectures mapped to specific playlists with status toggles (active/deactive).
- **Student Feedback Monitoring:** Review comments and reactions posted on uploaded tutorial videos.

2.6 Software Engineering Paradigm

The project adopts the **Iterative Waterfall Model**, which combines structured sequential development phases with iterative refinement loops.



Why This Paradigm is Chosen:

- **Phase Progression:** Each phase (SRS, database schema normalization, UI design, backend development) has clearly defined outputs.
- **Iterative Refinement:** Provides feedback loops between testing and coding to refine input sanitization filters, verify relational foreign key constraints, and enhance frontend responsiveness.
- **Maintainability:** Modular architecture ensures that backend database services remain decoupled from the frontend presentation layer.

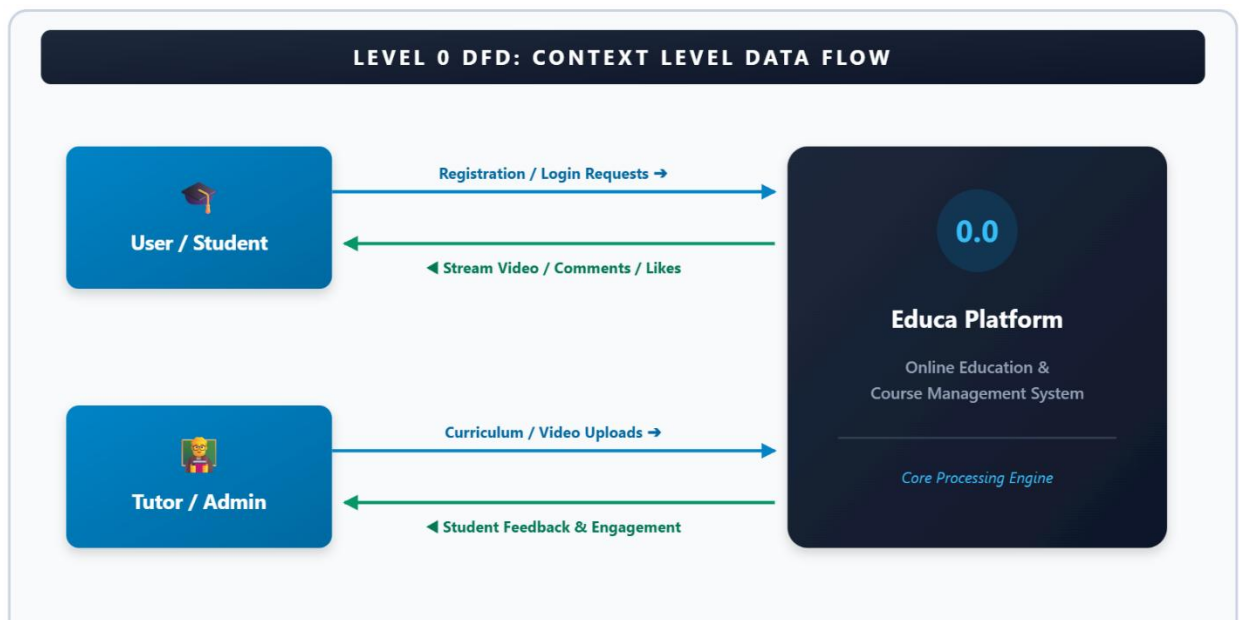
2.7 Data Flow Diagram:

A Data Flow Diagram (DFD) is a graphical representation of the flow of data through an information system, illustrating how data enters the system, gets transformed by various processes, and routes to data stores or external destinations.

Core Components of a DFD

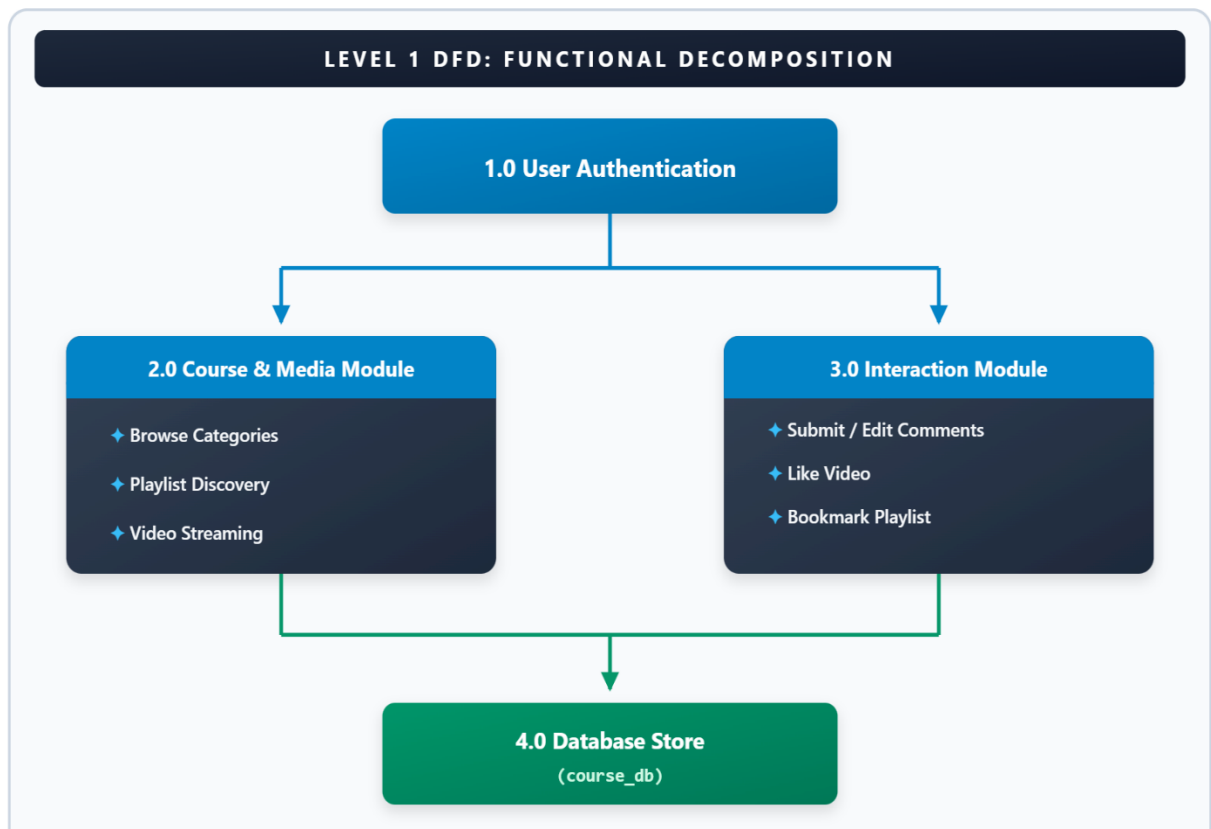
- **External Entity (Source / Sink):** Represents outside entities that interact with the system by providing inputs or receiving outputs (e.g., *User/Student*, *Tutor/Admin*). Usually drawn as a rectangle.
- **Process:** An activity or function that transforms incoming data into outgoing data (e.g., *Authenticate User*, *Manage Playlists*, *Post Comment*). Usually drawn as a circle or rounded rectangle.
- **Data Flow:** Depicts the path and direction that data travels between processes, data stores, and external entities. Represented as directional arrows labeled with the data being transferred.
- **Data Store:** Represents a repository or database table where data is stored for later retrieval (e.g., *course_db*, *users*, *playlist*, *content*). Represented as two parallel horizontal lines or an open rectangle.

Level 0 DFD (Context Level)



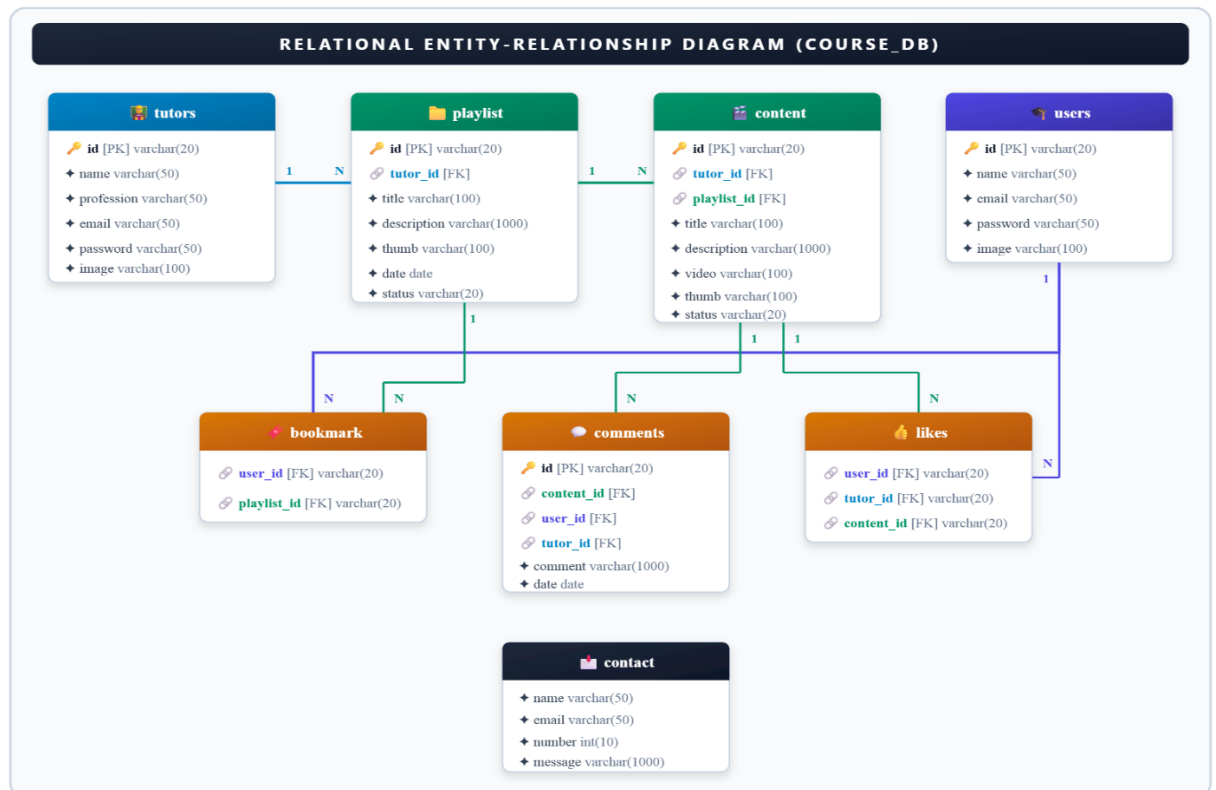
- Provides an overarching view of the entire system as a single central process connected to external entities.
- Establishes the boundaries and scope of the application without detailing internal subprocesses.

Level 1 DFD



- Decomposes the single Level 0 process into core functional modules (e.g., *Authentication Module*, *Course Management Module*, *Interaction Module*).
- Shows the primary data stores and how information moves between individual sub-systems.

ER Diagram



- **Strong Entities:** users, tutors, playlist, and content represent primary tables with dedicated primary key columns.
- **Associative / Weak Entities:** bookmark and likes act as junction bridge tables supporting many-to-many relationships across users and educational media

3. System Design & Database Structures

3.1. Module Specifications

- **1. User Module (Learners / Students):**
 - Authentication & Profiles: Register and authenticate via email and SHA-1 encrypted passwords with persistent cookie sessions (user_id).
 - Course Discovery: Browse the full course catalog, filter lectures by domain categories (Development, Business, Design, Science, etc.), and search topics.
 - Lecture Streaming: Watch video content inside an embedded player and view organized video series.
 - Social Engagement: Add, edit, and delete personal comments, like individual video lectures, and bookmark entire playlists to a personal dashboard.
- **2. Admin Module (Master Administrator):**
 - Platform Oversight: Monitor registered tutors, active users, uploaded course playlists, and total media content across the portal.
 - Content Governance: Review, activate, deactivate, or remove non-compliant course playlists and lecture videos.
 - Inquiry Management: Access, review, and manage visitor messages submitted through the contact interface (contact.php).
- **3. Sub-Admin Module (Tutors / Instructors):**
 - Instructor Authentication: Register and manage instructor profiles via dedicated tutor routes (admin/register.php / admin/login.php).
 - Curriculum Management: Author new playlists with custom titles, descriptions, and thumbnail images, and control visibility status (active/deactive).
 - Video Management: Upload MP4 video lectures mapped directly to specific course playlists with custom video posters.
 - Feedback Monitoring: View student comments and lecture likes across authored tutorials.

3.2 Data Structure of All Modules (course_db)

The system manages application state and relational records using 8 MySQL database tables:

Table: users

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 name	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 password	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 image	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

Stores registered student account details, profile picture file names, and authentication credentials.

Table: tutors

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 name	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 profession	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 password	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 image	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More

Stores instructor credentials, bios, professional titles, and profile images.

Table: playlist

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 tutor_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 title	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 description	varchar(1000)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 thumb	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 date	date			No	current_timestamp()			Change Drop More
☐	7 status	varchar(20)	utf8mb4_general_ci		No	deactive			Change Drop More

Stores uploaded video files, associated playlist references, and descriptions for individual lectures.

Table: bookmark

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 user_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 playlist_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More

Associative junction table linking users to their saved course playlists.

Table: likes

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 user_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 tutor_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 content_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More

Associative junction table recording student likes on specific video lectures.

Table: contact

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 name	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 email	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 number	int(10)			No	None			Change Drop More
☐	4 message	varchar(1000)	utf8mb4_general_ci		No	None			Change Drop More

Stores visitor support inquiries submitted via the platform's contact form.

Table: comments

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 content_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 user_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 tutor_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 comment	varchar(1000)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 date	date			No	current_timestamp()			Change Drop More

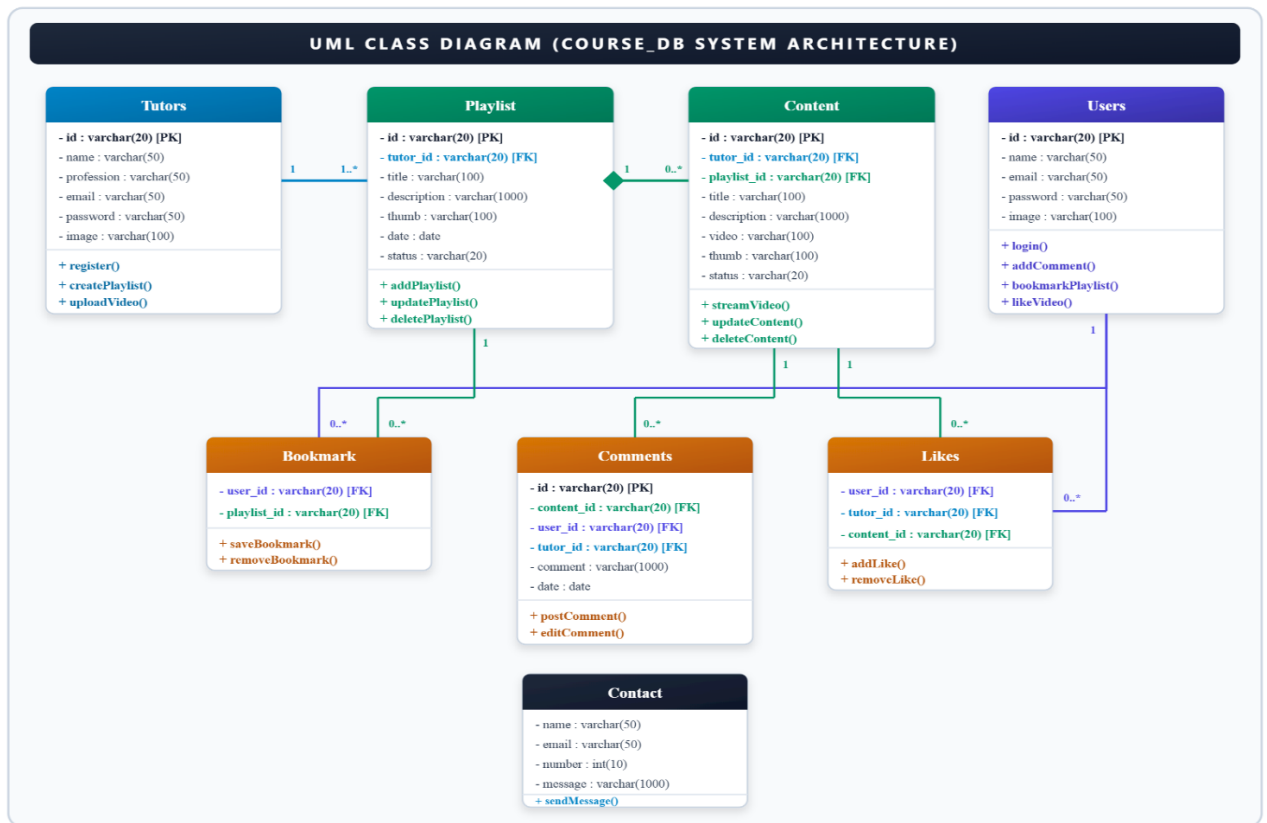
Manages interactive user feedback and discussions on specific video lessons.

Table: content

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	2 tutor_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	3 playlist_id	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More
☐	4 title	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	5 description	varchar(1000)	utf8mb4_general_ci		No	None			Change Drop More
☐	6 video	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	7 thumb	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
☐	8 date	date			No	current_timestamp()			Change Drop More
☐	9 status	varchar(20)	utf8mb4_general_ci		No	deactive			Change Drop More

Stores metadata and media paths for individual video lectures uploaded by instructors.

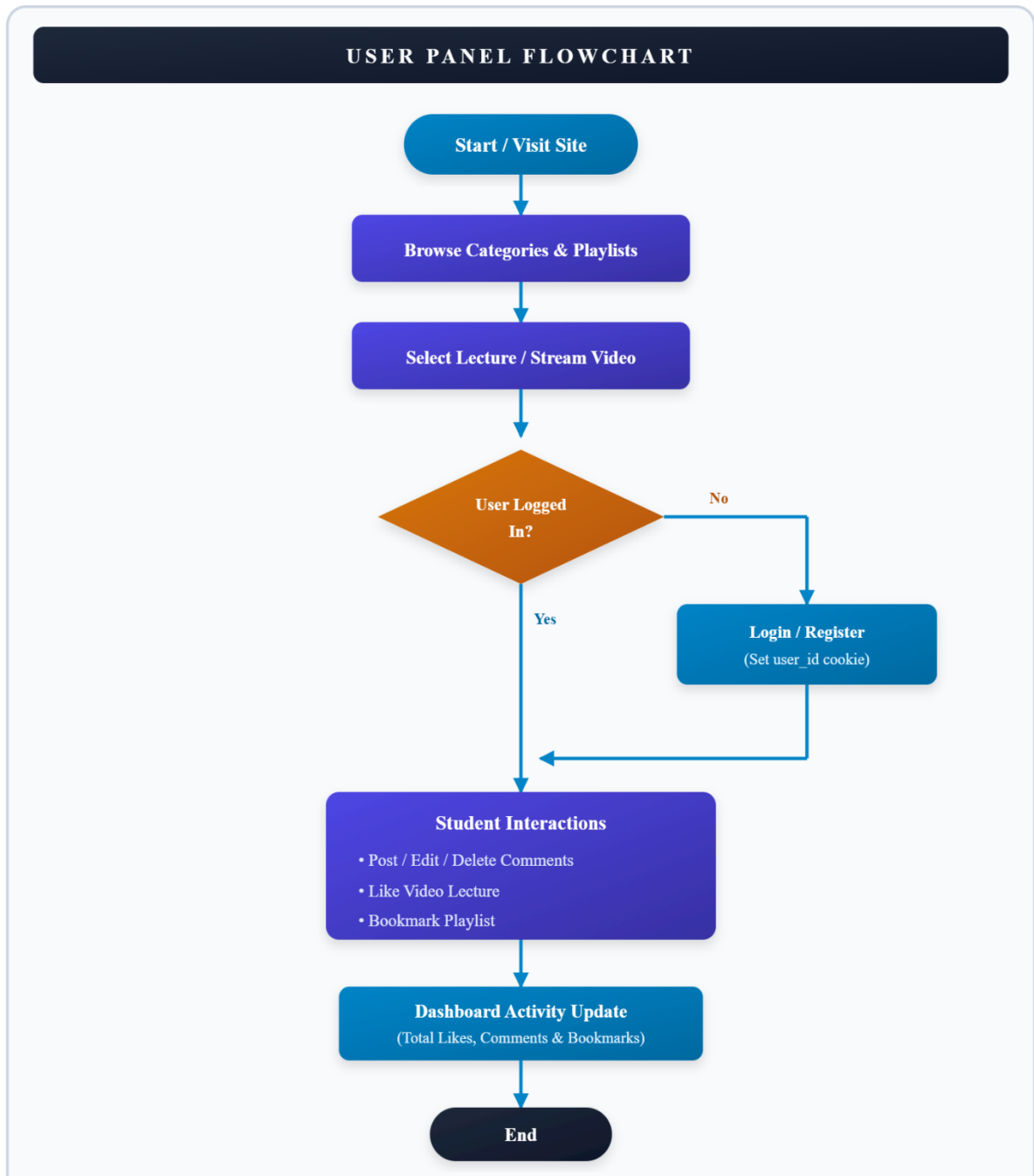
Relationship between tables (Class Diagram)



3.3 PROCEDURAL DESIGN:

3.3.1 User Panel Design

In the user panel design, all interactions and facilities are structured for student and learner workflows. The public and registered interfaces allow visitors to browse the course catalog, explore categorized topics, stream instructional video lectures, and participate in discussion threads.



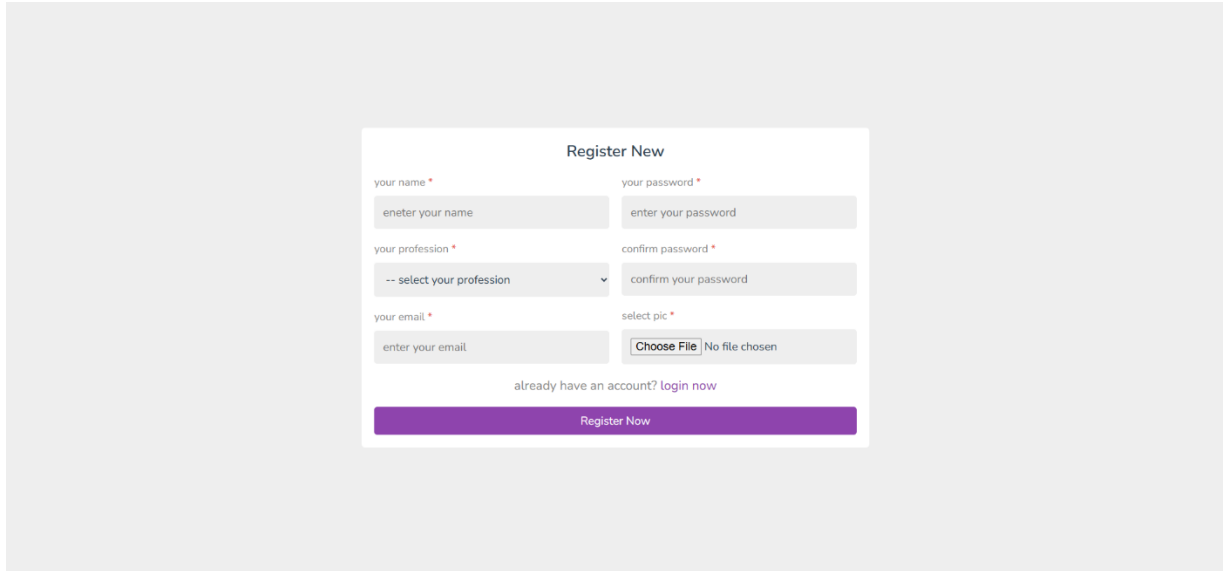
3.3.2 Admin Panel Design

The Admin and Tutor panel in the Educa platform is structured to provide instructors and administrators with an isolated, secure control environment for managing curriculum, media files, and user interactions.



4.SCREENSHOT

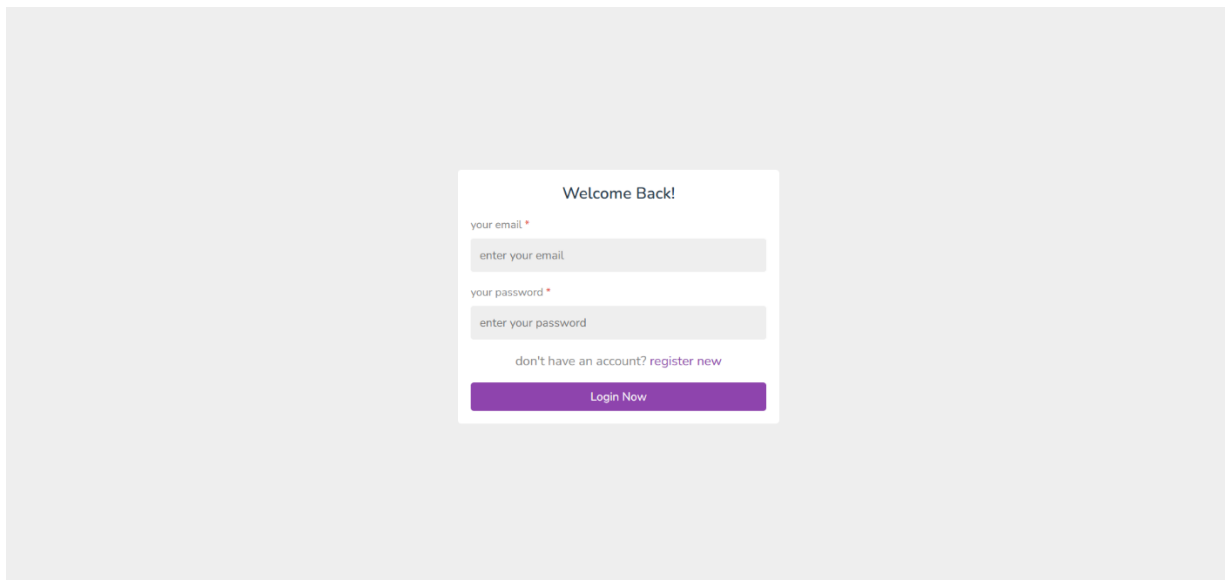
Admin Register



The screenshot shows a 'Register New' form with the following fields and elements:

- your name ***: Input field with placeholder 'enter your name'.
- your password ***: Input field with placeholder 'enter your password'.
- your profession ***: Dropdown menu with placeholder '-- select your profession'.
- confirm password ***: Input field with placeholder 'confirm your password'.
- your email ***: Input field with placeholder 'enter your email'.
- select pic ***: File upload area with a 'Choose File' button and 'No file chosen' text.
- already have an account? login now**: Link text.
- Register Now**: Purple button.

Admin Login



The screenshot shows a 'Welcome Back!' login form with the following fields and elements:

- your email ***: Input field with placeholder 'enter your email'.
- your password ***: Input field with placeholder 'enter your password'.
- don't have an account? register new**: Link text.
- Login Now**: Purple button.

Admin Dashboard



Mohan Panthagani
developer

View Profile

home

playlists

contents

comments

logout

Admin.

Dashboard

welcome!

Mohan Panthagani

View Profile

5

total contents

Add New Content

2

total playlists

Add New Playlist

1

total likes

View Contents

2

total comments

View Comments

quick select

login or register

LoginRegister

Warning: include(components/footer.php): Failed to open stream: No such file or directory in C:\xampp\htdocs\Project\admin\dashboard.php on line 112

Warning: include(): Failed opening 'components/footer.php' for inclusion (include_path='C:\xampp\htdocs\PEAR') in C:\xampp\htdocs\Project\admin\dashboard.php on line 112

Admin Add Playlists



Mohan Panthagani
developer

View Profile

home

playlists

contents

comments

logout

Admin.

Added Playlists

create new playlist

Add Playlist

active

2026-08-27

4#01

Complete HTML Tutorial

An HTML description refers either to the semantic description list element used for term-definition...

UpdateDelete

View Playlist

active

2026-08-27

1#01

Complete CSS tutorial

CSS (Cascading Style Sheets) is the standard language used to control the visual presentation of web...

UpdateDelete

View Playlist

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21

Admin Add Content



Mohan Panthagani
developer

View Profile

home

playlists

contents

comments

logout

Admin.

Your Contents

create new content

Add Content

active 2026-08-27

#01

HTML

Update Delete

View Content

active 2026-08-27

#02

HTML

Update Delete

View Content

active 2026-08-27

#03

active 2026-08-27

#04

active 2026-08-27

#01

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Admin Profile



Mohan Panthagani
developer

View Profile

home

playlists

contents

comments

logout

Admin.

Profile Details



Mohan Panthagani
developer

Update Profile

2

total playlists

View Playlists

5

total videos

View Contents

1

total likes

View Contents

2

total comments

View Comments

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Admin Profile Update



Mohan Panthagani
developer

View Profile

home

playlists

contents

comments

logout

Admin.

Update Profile

your name

Mohan Panthagani

your profession

developer

your email

mohan@gmail.com

update pic :

Choose File No file chosen

old password :

enter your old password

new password :

enter your new password

confirm password :

confirm your new password

Update Now

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User Register

please login or register

LoginRegister

home

about us

courses

teachers

contact us

Educa.

search courses...

Create Account

your name *

enter your name

your password *

enter your password

your email *

enter your email

confirm password *

confirm your password

select pic *

Choose FileNo file chosen

already have an account? login now

Register Now

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User Login

please login or register

LoginRegister

home

about us

courses

teachers

contact us

Educa.

search courses...

Welcome Back!

your email *

enter your email

your password *

enter your password

don't have an account? register now

Login Now

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User Dashboard

Mohan Panthagani

student

View Profile

home

about us

courses

teachers

contact us

Educa.

search courses...

Quick Options

likes and comments

total likes : 1

View Likes

total comments : 2

View Comments

saved playlist : 1

View Bookmark

top categories

development

business

marketing

photography

software

design

music

science

popular topics

HTML

javascript

PHP

CSS

react

bootstrap

become a tutor

Lorem ipsum dolor sit amet
consectetur adipisicing elit.
Ipsa, laudantium.

Get Started

Latest Courses

Mohan Panthagani

2026-08-27

#01

Mohan Panthagani

2026-08-27

#01

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ffjifsidff

23

User view Playlist



Mohan Panthagani
student

View Profile

- home
- about us
- courses
- teachers
- contact us

Educa.

search courses...

All Courses



Mohan Panthagani
2026-08-27

#01

Complete HTML Tutorial

View Playlist



Mohan Panthagani
2026-08-27

#01

Complete CSS tutorial

View Playlist

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User Teachers



Mohan Panthagani
student

View Profile

- home
- about us
- courses
- teachers
- contact us

Educa.

search courses...

Expert Tutors

search tutor...

become a tutor

Lorem ipsum dolor sit, amet consectetur adipiscing elit. Laborum, magnam!

Get Started



Mohan Panthagani
developer

playlists : 2
total videos : 5
total likes : 1
total comments : 2

View Profile



Mohan
engineer

playlists : 0
total videos : 0
total likes : 0
total comments : 0

View Profile

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User Contact Us



Mohan Panthagani
student

View Profile

- home
- about us
- courses
- teachers
- contact us

Educa.

search courses...



get in touch

enter your name

enter your email

enter your number

enter your message

Send Message

phone number

123-456-7890
111-222-3333

email address

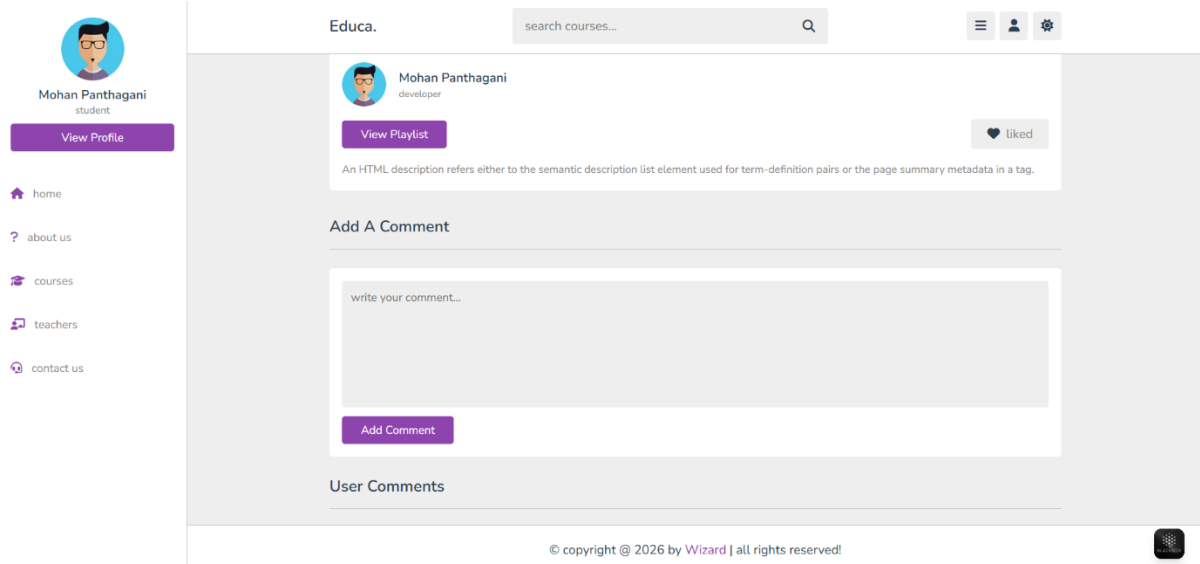
shaikhana@gmail.com
anashai@mail.ru

office address

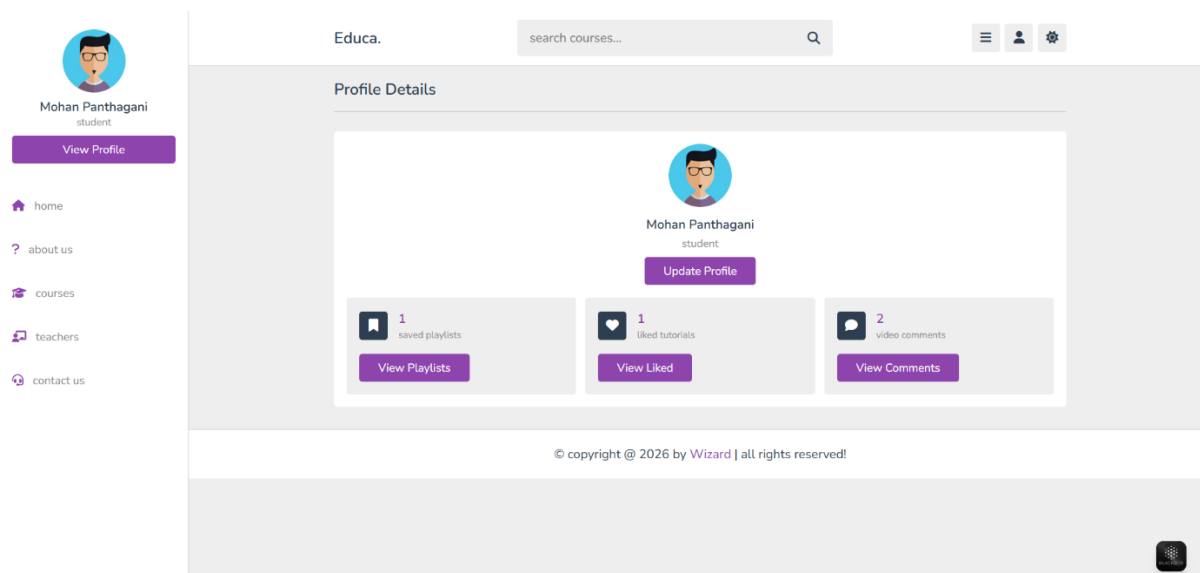
flat no. 1, a-1 building, jogeshwari, mumbai,
india - 400104

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User Add Comments



User Profile



5.Coding

Login.PHP

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Admin Login</title>

    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.2.0/css/all.min.css">

    <link rel="stylesheet" href="{ { url_for('static', filename='css/admin_style.css') } }">

</head>

<body style="padding-left: 0;">

    {% with messages = get_flashed_messages() %}

        {% if messages %}

            {% for msg in messages %}

                <div class="message form">

                    <span>{{ msg }}</span>

                    <i class="fas fa-times" onclick="this.parentElement.remove();"></i>

                </div>

            {% endfor %}

        {% endif %}

    {% endwith %}

    <section class="form-container">

        <form action="" method="post" class="login">

            <h3>welcome back!</h3>

            <p>your email <span>*</span></p>

            <input type="email" name="email" placeholder="enter your email" maxlength="50" required class="box">

            <p>your password <span>*</span></p>

            <input type="password" name="pass" placeholder="enter your password" maxlength="20" required class="box">
```

```
<p class="link">don't have an account? <a href="{{ url_for('admin.register') }}">register  
new</a></p>
```

```
<input type="submit" name="submit" value="login now" class="btn">  
</form>  
</section>
```

```
<script>  
let darkMode = localStorage.getItem('dark-mode');  
let body = document.body;  
if(darkMode === 'enabled'){ body.classList.add('dark'); }  
</script>  
</body>  
</html>
```

Profile.PHP

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-  
scale=1.0">  
  <title>Profile</title>  
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-  
awesome/6.2.0/css/all.min.css">  
  <link rel="stylesheet" href="{{ url_for('static', filename='css/admin_style.css') }}">  
</head>  
<body>  
  {% include 'components/admin_header.html' %}  
  
  <section class="tutor-profile" style="min-height: calc(100vh - 19rem);">  
    <h1 class="heading">profile details</h1>  
    <div class="details">  
      <div class="tutor">  
          
        <h3>{{ tutor.name }}</h3>  
        <span>{{ tutor.profession }}</span>
```

```

        <a href="{{ url_for('admin.update_profile') }}" class="inline-btn">update
profile</a>

    </div>

    <div class="flex">

        <div class="box">

            <span>{{ total_playlists }}</span>

            <p>total playlists</p>

            <a href="{{ url_for('admin.playlists') }}" class="btn">view playlists</a>

        </div>

        <div class="box">

            <span>{{ total_contents }}</span>

            <p>total videos</p>

            <a href="{{ url_for('admin.contents') }}" class="btn">view contents</a>

        </div>

        <div class="box">

            <span>{{ total_likes }}</span>

            <p>total likes</p>

            <a href="{{ url_for('admin.contents') }}" class="btn">view contents</a>

        </div>

        <div class="box">

            <span>{{ total_comments }}</span>

            <p>total comments</p>

            <a href="{{ url_for('admin.comments') }}" class="btn">view comments</a>

        </div>

    </div>

</div>

</div>

</section>

{% include 'components/footer.html' %}

<script src="{{ url_for('static', filename='js/admin_script.js') }}"></script>

</body>

</html>

```

6. Testing

Testing for the Online Education & Course Management System (Educa) follows a progressive hierarchy, verifying individual components before validating complete end-to-end workflows.

1. Unit Testing

- **ID Generation Routine (unique_id()):** Evaluated the custom function to ensure the generated 20-character alphanumeric strings are randomized and collision-free.
- **Input Data Sanitization:** Checked form fields across registration, comments, and contact scripts to confirm that `filter_var(..., FILTER_SANITIZE_STRING)` strips malicious scripts, special characters, and tags.

2. Module Testing

- **User Authentication Module:** Tested user registration, cookie assignment (`user_id`), profile picture uploads, and session termination via `logout.php`.
- **Course & Media Module:** Verified queries fetching courses by category and validated status filters ensuring deactivated content (`status = 'deactive'`) is hidden from students.
- **Interactive Feedback Module:** Checked that users can post comments, edit their own posts, toggle video likes, and bookmark course playlists.
- **Tutor/Admin Control Module:** Verified instructor authentication (`admin/login.php`), playlist creation with thumbnails, and MP4 video uploads.

3. Sub-system Testing

- **User-Engagement Sub-system:** Verified the interaction pipeline connecting video playback (`watch_video.php`), the discussion stream (`comments.php`), and student metrics on the user dashboard.
- **Curriculum Management Sub-system:** Tested communication between the instructor dashboard, database storage (playlist and content tables), and the file system directory (`../uploaded_files/`).
- **Database & Security Sub-system:** Validated that PHP Data Objects (PDO) prepared statements properly bind parameters across all database queries to prevent SQL injection vulnerabilities.

4. System Testing

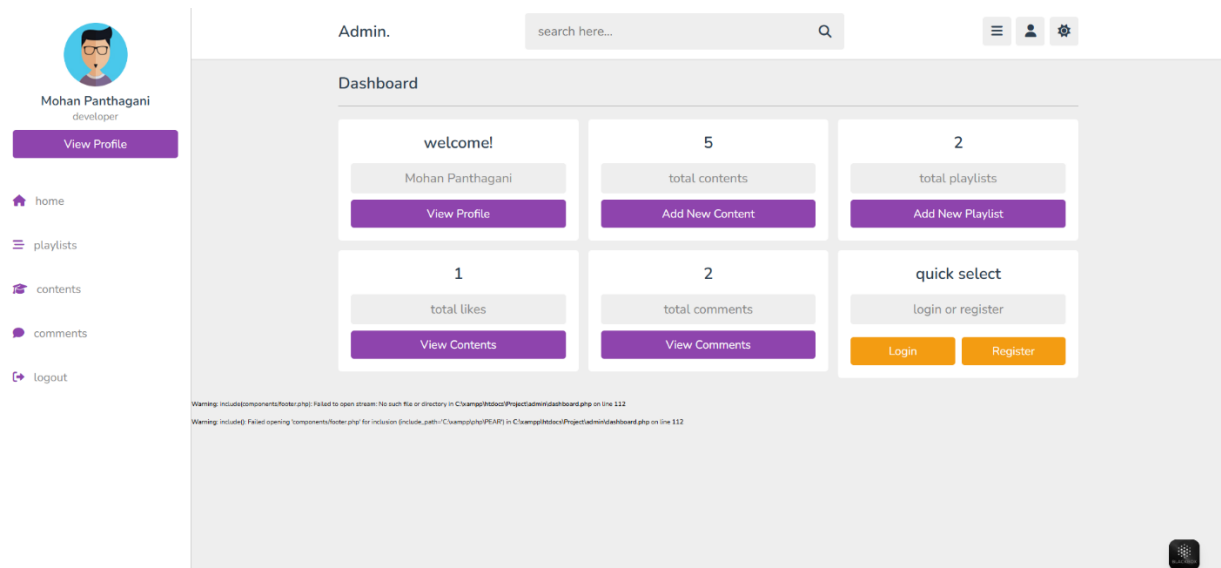
- **End-to-End Workflow Execution:** Tested complete user journeys—from student registration and course exploration to video streaming, commenting, and bookmarking.
- **Cross-Browser & Platform Compatibility:** Tested the interface on multiple web browsers (Google Chrome, Microsoft Edge, Mozilla Firefox) to confirm consistent styling, video playback, and layout responsiveness.

5. Acceptance Testing (User Acceptance Testing - UAT)

- **Student Role Validation:** Learners verified that navigating topics, searching courses, saving bookmarks, and interacting with videos is intuitive and responsive.
- **Instructor Role Validation:** Tutors confirmed that creating course playlists, uploading video lessons, and moderating student comments operates without technical friction.
- **Error Handling & Feedback:** Verified that user-facing alerts (such as *"image size is too large!"*, *"comment deleted successfully!"*, or *"incorrect email or password!"*) provide clear and accurate guidance.

Report

Admin dashboard Report



7. Future scope

1. Advanced Learning & Assessments

- **Quizzes & Assignments:** Integrate automated MCQ-based tests, downloadable resources (PDFs, notes), and assignment submission portals for each video lecture.
- **Certificate Generation:** Implement automated PDF course completion certificates with verifiable unique IDs.

2. Monetization & Commerce

- **Payment Gateway Integration:** Incorporate secure checkout gateways (Stripe, Razorpay, PayPal) for premium/paid courses and subscription plans.

3. Enhanced Communication & Interaction

- **Live Classes & Webinars:** Integrate Google Meet or Zoom API for real-time live streaming, screen sharing, and virtual classroom sessions.

4. Infrastructure & Security Enhancements

- **Cloud Storage & CDN Delivery:** Migrate local video storage to cloud services (AWS S3, Cloudflare Stream) with adaptive bitrate streaming (HLS) for faster buffering.
- **Mobile Application Development:** Build cross-platform mobile apps (React Native or Flutter) with offline video download capabilities.

Conclusion

The Online Education & Course Management System (Educa) was designed and developed to provide a lightweight, scalable, and responsive web platform for digital course distribution and student-instructor collaboration. By following a structured software development lifecycle, the system addresses core challenges in organizing technical video content and facilitating structured online learning.

The project fulfilled all functional and non-functional requirements outlined in the Software Requirement Specification (SRS). Through multi-level verification—ranging from unit and module testing to complete User Acceptance Testing (UAT)—the system demonstrated strong stability, secure session management, and efficient media delivery. The platform provides a robust foundation that can be expanded with cloud media storage, automated certification, and live classroom integration in future iterations.

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